



Inspiring Excellence

CSE422: Artificial Intelligence

Project Name: Predicting Movie Rating Categories using
Machine Learning and Neural Networks

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1. Introduction:

This project focuses on building machine learning models to predict movie rating categories based on various features like budget, director popularity, actor popularity, number of awards, and more. The goal is to help movie producers or platforms like IMDb forecast the likely rating category (e.g., Excellent, Good, Average) of a movie prior to release, supporting strategic decision-making in production and marketing.

Target: Predicting Movie Rating Categories using Machine Learning and Neural Networks.

2. Dataset Description:

1. **Source:** [movie_ratings_dataset.csv](#)
2. **Number of Data Points:** 1200
3. **Number of Features:** 11 independent + 1 target = 12 total
4. **Target Variable:** Rating_Category
5. **Type of Problem:** Classification (target is categorical)
6. **Feature Types:**
 - Quantitative:** Director_Popularity, Budget_MillionUSD, Runtime_Minutes, Num_Awards_Won, etc.

Categorical: Genre, Is_Sequel, Has_Famous_Producer

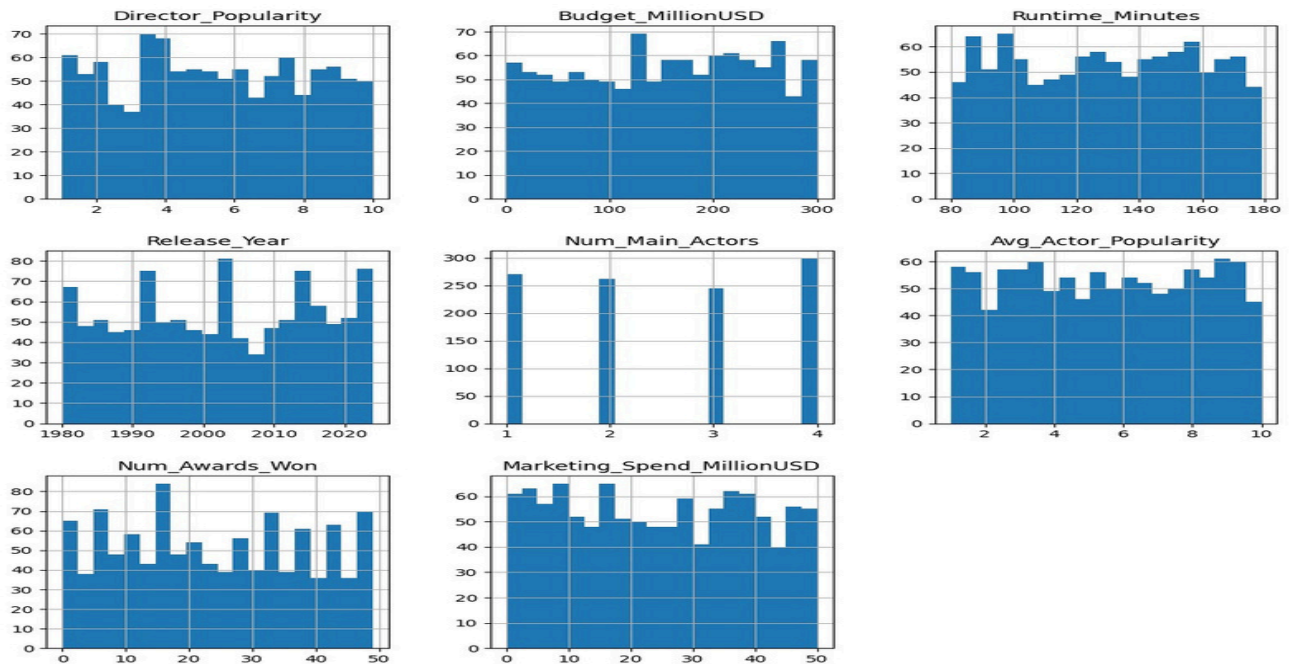


Figure: Histogram of all the Categorical Features

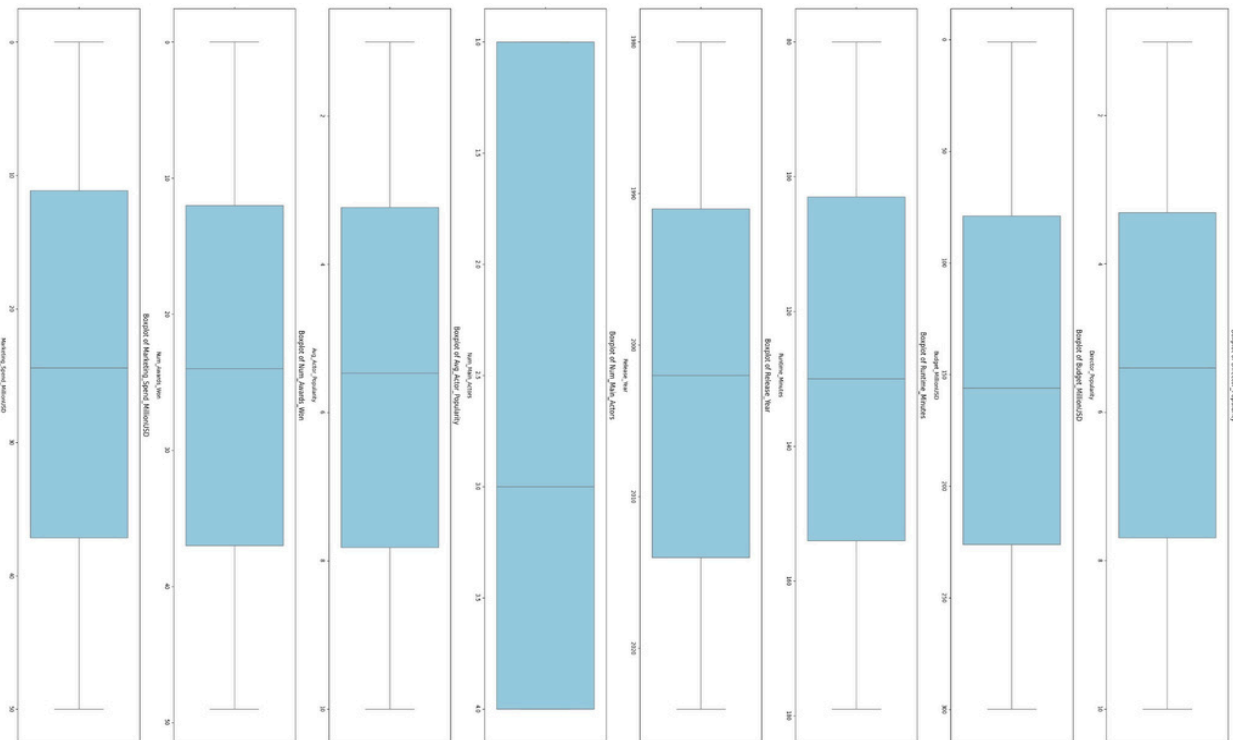


Figure: BoxPlot of all the Numerical Features

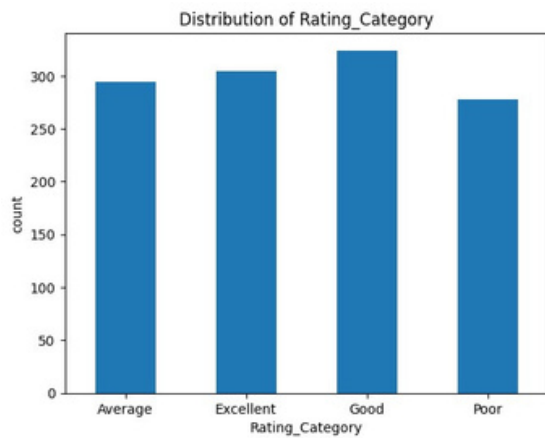


Figure: Rating Category

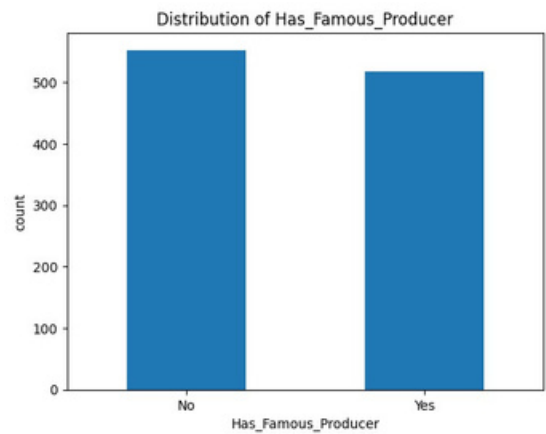


Fig: Famous Producer

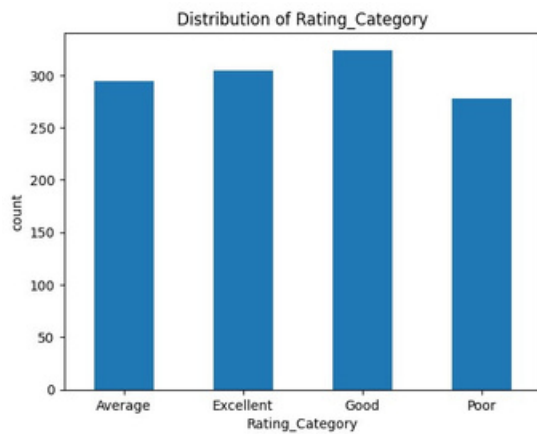


Figure: Rating Category

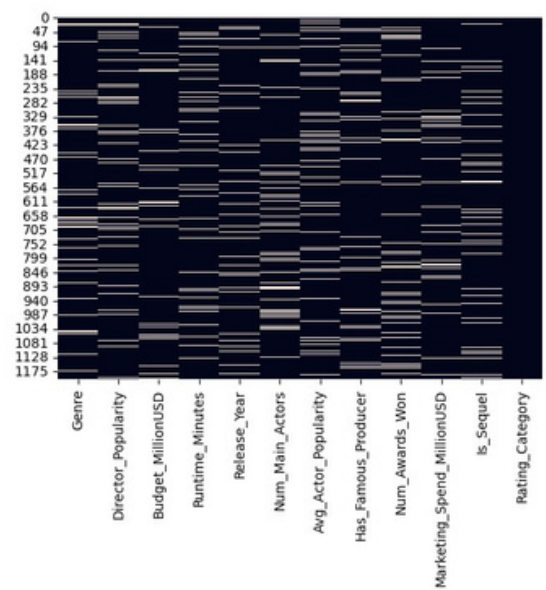


Figure: Heat Map for Null Values

3. Imbalanced Dataset Analysis

A bar chart of the `Rating_Category` column revealed that the dataset is moderately imbalanced. For example, “Excellent” had more instances than “Average.” This was considered while choosing metrics and evaluation strategies.



4. Exploratory Data Analysis (EDA)

Key EDA findings:

1. Higher budgets and actor popularity correlate with higher rating categories.
2. Sequels and famous producers often appeared in high-rated movies.

3. Runtime and release year showed weak correlations with the rating category

Genre	Very low with all
Budget_MillionUSD	Low with all
Release_Year	Low with all
Avg_Actor_Popularity	Low with all
Num_Awards_Won	Low with all
Is_Seque	Low with all

5. Dataset Preprocessing

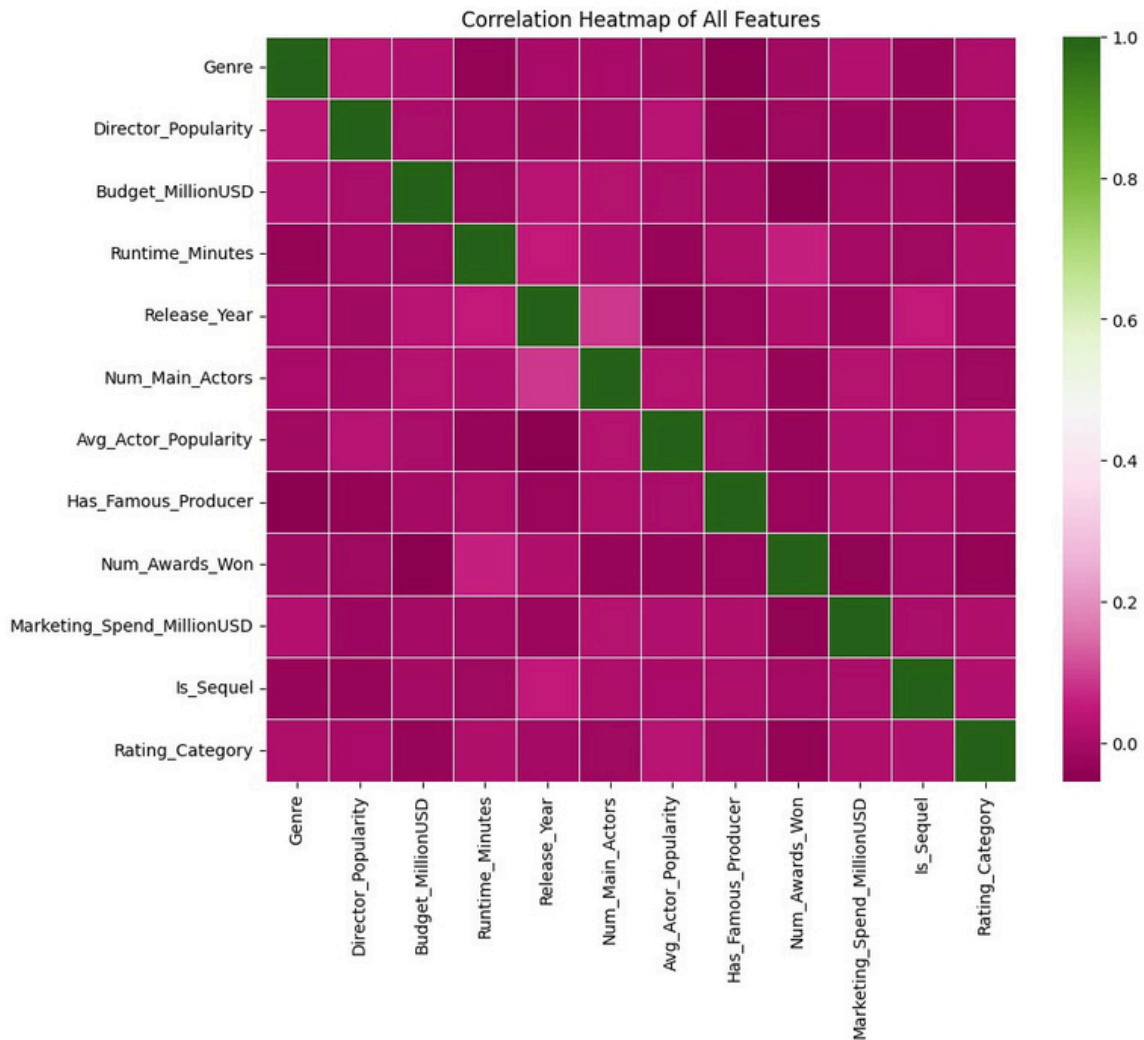
1. Missing Values:

(e.g., Budget_MillionUSD, Runtime_Minutes) were imputed using mean. Categorical columns (e.g., Is_Sequel, Genre) were filled using random sampling from valid categories. Release_Year were filled with random years (1950–2025) to preserve distribution.

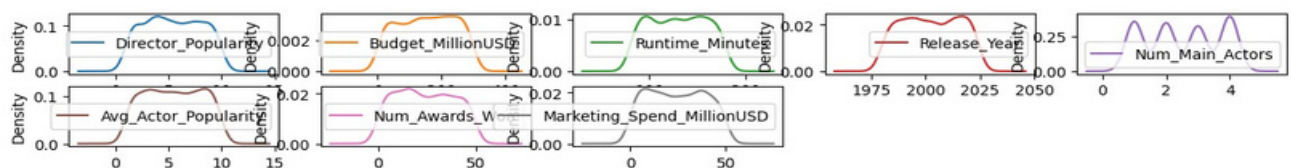
2. Categorical Encoding:

Genre manually mapped to integers (e.g., Action=1, Horror=2, etc.). Has_Famous_Producer, Is_Sequel were encoded as 0 or 1. Rating_Category (target) were mapped as: Good=1, Excellent=2, Average=3, Poor=4.

Correlation Heatmap: A correlation heatmap was created using seaborn to identify relationships between numerical features. Highly correlated features were checked for redundancy. It helped in prioritizing feature importance and understanding multicollinearity.



Density plot of Numerical features



6. Dataset Splitting

-Split into 70% training and 30% testing data used to maintain class balance across training and test sets.

```
1 #Training
2 x_train, x_test, y_train, y_test = train_test_split(X, y, test_size = 0.3, random_state = 60)
```

7. Model Training & Testing

The following models were retrained and tested:

- **Logistic Regression (Classification):**

```
1 #Logistic Regression
2 model = LogisticRegression()
3 model.fit(x_train, y_train)
4 y_pred = model.predict(x_test)
5
6 accuracy = accuracy_score(y_test, y_pred)
7 conf_matrix = confusion_matrix(y_test, y_pred)
8 report = classification_report(y_test, y_pred)
9
10 print("Accuracy:", accuracy)
11 print("Confusion Matrix:\n", conf_matrix)
12 print("Classification Report:\n", report)
```

- Decision Tree (Classification):

```
[105] 1 #DecisionTreeClassifier
      2 from sklearn.tree import DecisionTreeClassifier
      3 model = DecisionTreeClassifier(random_state=42)
      4 model.fit(x_train, y_train)
      5 y_pred = model.predict(x_test)
      6
      7 accuracy = accuracy_score(y_test, y_pred)
      8 conf_matrix = confusion_matrix(y_test, y_pred)
      9 report = classification_report(y_test, y_pred)
     10
     11 print("Accuracy:", accuracy)
     12 print("Confusion Matrix:\n", conf_matrix)
     13 print("Classification Report:\n", report)
```

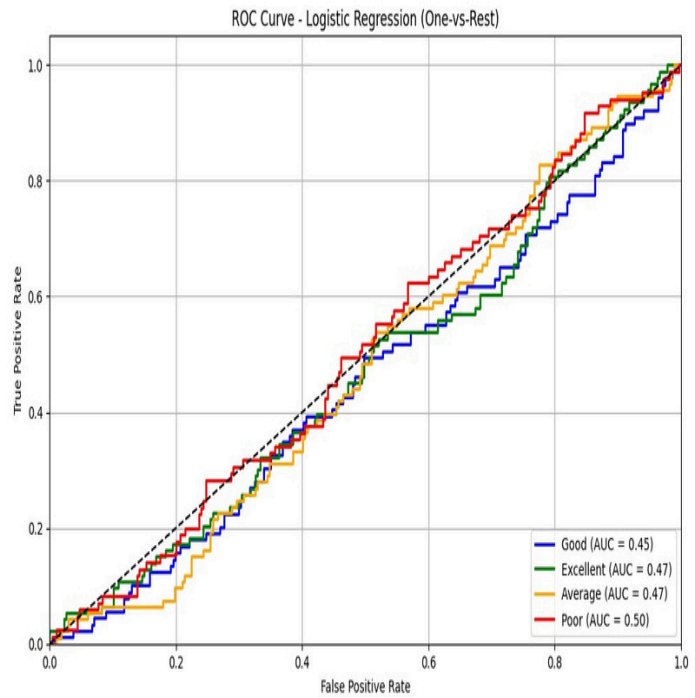
- Neural Network

```
1 #Neural Network
2
3 from sklearn.neural_network import MLPClassifier
4 from sklearn.metrics import accuracy_score, confusion_matrix, classifica
5 model = MLPClassifier(hidden_layer_sizes=(128,), activation='relu', solv
6                       max_iter=300, batch_size=128, random_state=42)
7
8 model.fit(x_train, y_train)
9 y_pred = model.predict(x_test)
10
11 accuracy = accuracy_score(y_test, y_pred)
12 conf_matrix = confusion_matrix(y_test, y_pred)
13 report = classification_report(y_test, y_pred)
14
15 print("Accuracy:", accuracy)
16 print("Confusion Matrix:\n", conf_matrix)
17 print("Classification Report:\n", report)
```

8. Model Evaluation & Comparison:

	Logistic Regression																											
Accuracy	0.23																											
Precision, recall, F-1 Score	0.17	0.23	0.17																									
Confusion Matrix	Confusion Matrix Heatmap <table><tr><th>True Label \ Predicted Label</th><th>Good</th><th>Excellent</th><th>Average</th><th>Poor</th></tr><tr><th>Good</th><td>55</td><td>27</td><td>0</td><td>7</td></tr><tr><th>Excellent</th><td>63</td><td>21</td><td>0</td><td>9</td></tr><tr><th>Average</th><td>64</td><td>18</td><td>0</td><td>11</td></tr><tr><th>Poor</th><td>61</td><td>16</td><td>1</td><td>7</td></tr></table>			True Label \ Predicted Label	Good	Excellent	Average	Poor	Good	55	27	0	7	Excellent	63	21	0	9	Average	64	18	0	11	Poor	61	16	1	7
True Label \ Predicted Label	Good	Excellent	Average	Poor																								
Good	55	27	0	7																								
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Multiclass ROC AUC Score (0vR)	0.50																											

ROC Curve

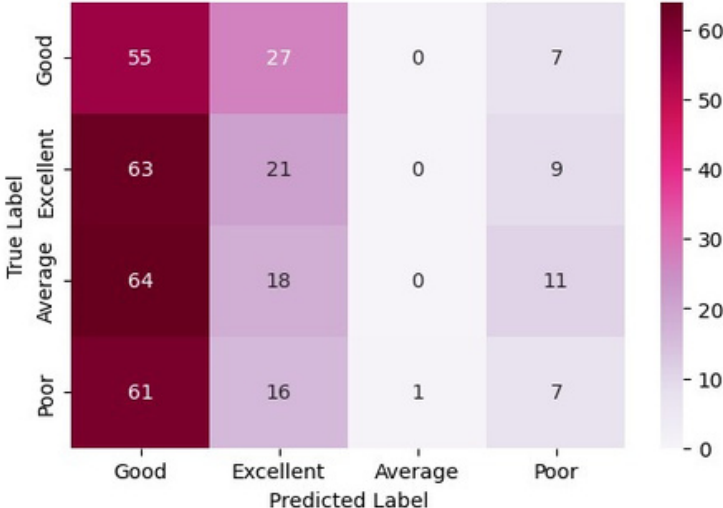
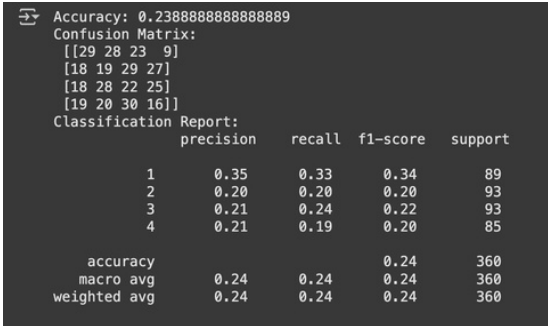


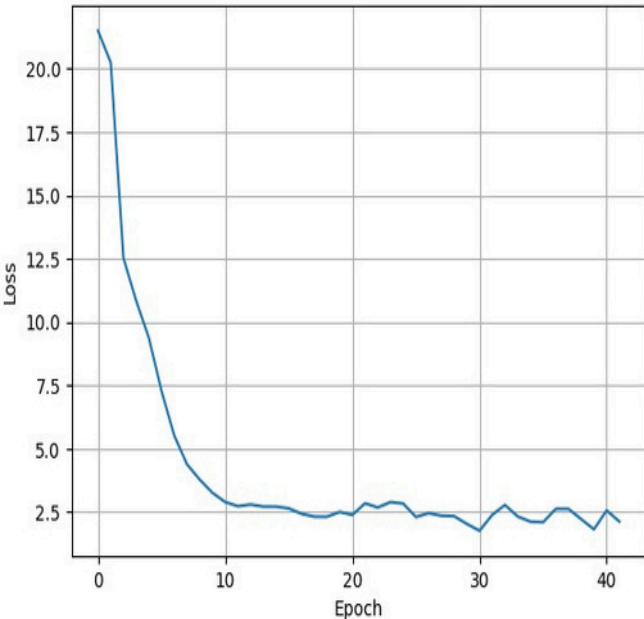
Screenshot

```
Accuracy: 0.2305555555555557
Confusion Matrix:
[[55 27  0  7]
 [63 21  0  9]
 [64 18  0 11]
 [61 16  1  7]]
Classification Report:
      precision    recall  f1-score   support

     1       0.23       0.62       0.33        89
     2       0.26       0.23       0.24        93
     3       0.00       0.00       0.00        93
     4       0.21       0.08       0.12        85

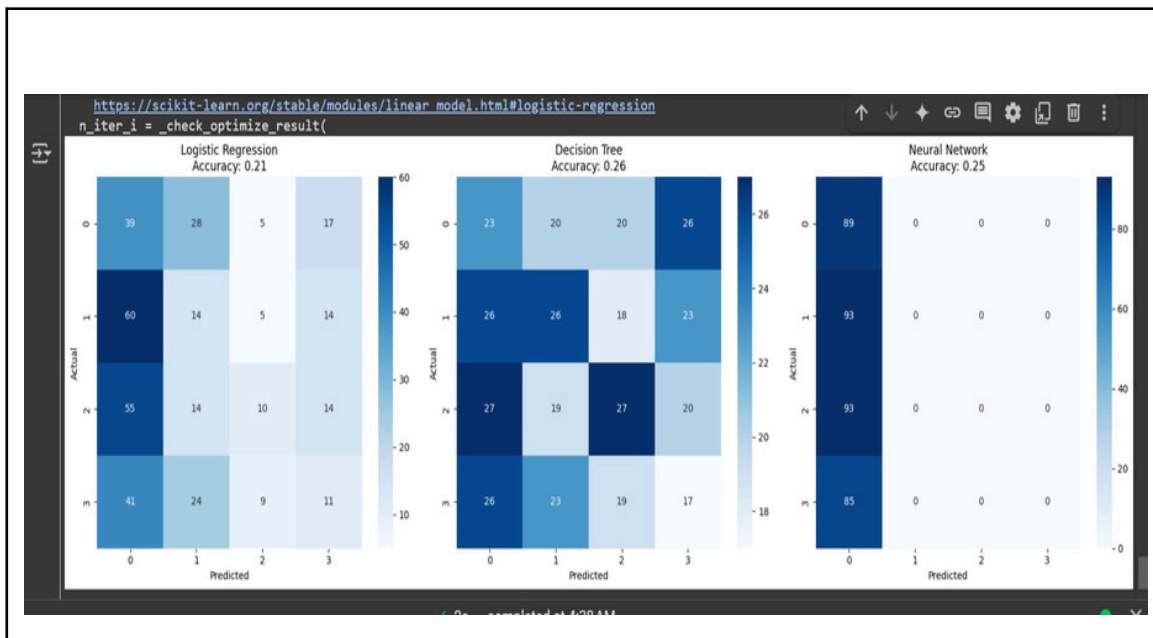
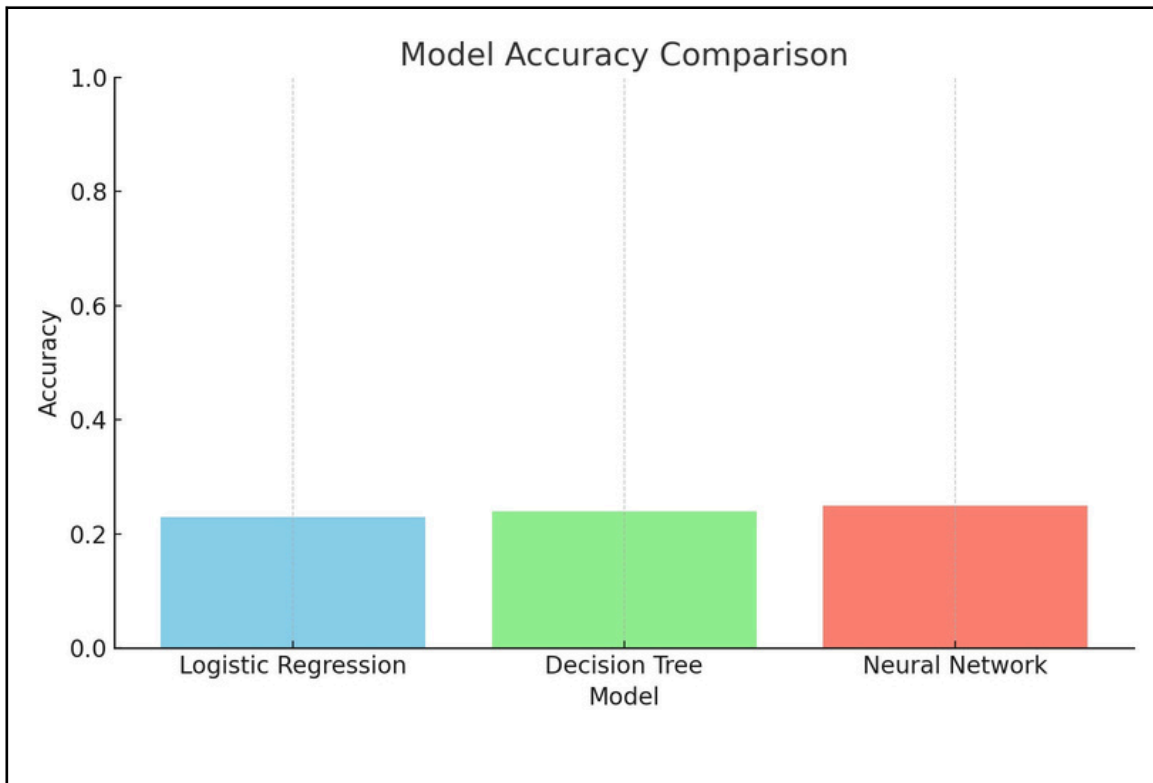
 accuracy          0.23        360
 macro avg         0.17        0.23        0.17        360
 weighted avg      0.17        0.23        0.17        360
```

	Decision Tree																														
Accuracy	0.24																														
Confusion Matrix	Confusion Matrix Heatmap  <table><tr><td>True Label</td><td>Good</td><td>Excellent</td><td>Average</td><td>Poor</td></tr><tr><td>Good</td><td>55</td><td>27</td><td>0</td><td>7</td></tr><tr><td>Excellent</td><td>63</td><td>21</td><td>0</td><td>9</td></tr><tr><td>Average</td><td>64</td><td>18</td><td>0</td><td>11</td></tr><tr><td>Poor</td><td>61</td><td>16</td><td>1</td><td>7</td></tr><tr><td></td><td>Good</td><td>Excellent</td><td>Average</td><td>Poor</td></tr></table> Predicted Label	True Label	Good	Excellent	Average	Poor	Good	55	27	0	7	Excellent	63	21	0	9	Average	64	18	0	11	Poor	61	16	1	7		Good	Excellent	Average	Poor
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	Good	Excellent	Average	Poor																											
Precision, Recall, F1 Score	0.24 0.24 0.24																														
ROC AUC Score (0vR)	0.49																														
Screenshot																															

	Neural Network			
Accuracy	0.25			
Confusion Matrix				
Precision, Recall, F1 Score	0.06	0.25	0.10	
Loss Curve	Loss Curve 			
Screenshot	<pre>Accuracy: 0.2472222222222223 Confusion Matrix: [[89 0 0 0] [93 0 0 0] [93 0 0 0] [85 0 0 0]] Classification Report: precision recall f1-score support 1 0.25 1.00 0.40 89 2 0.00 0.00 0.00 93 3 0.00 0.00 0.00 93 4 0.00 0.00 0.00 85 accuracy macro avg 0.06 0.25 0.10 360 weighted avg 0.06 0.25 0.10 360</pre>			

Model Accuracy Comparison:

Best Model is the Neural Network by a slim margin.



9. Conclusion

The results indicate that all models performed poorly, with the neural network achieving the highest accuracy at just 0.25. This suggests weak predictive signals in the data and minimal correlation between features and the target variable.

Contributing factors include a relatively small dataset, label encoding issues, class imbalance, and random imputation, which may have introduced noise. Key challenges were managing missing data, preprocessing inconsistencies, and tuning the neural network. Despite low performance, the project reinforced the importance of high-quality, balanced data and thoughtful feature selection in model development.